

241186_4_20032026

March 27, 2026

Question 7

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[ ]: from google.colab import files

print("Please upload the 'Q2_data.csv' file:")
files.upload()
```

Please upload the 'Q2_data.csv' file:

<IPython.core.display.HTML object>

Saving A4_test_a4d66d34-6db0-44a4-a7c1-0aa26ab8d705.csv to

A4_test_a4d66d34-6db0-44a4-a7c1-0aa26ab8d705 (1).csv

Saving A4_train_74e601c5-f446-4f34-a53b-28ce05d1cec4.csv to

A4_train_74e601c5-f446-4f34-a53b-28ce05d1cec4 (1).csv

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1.9038456459141684,1.687404860766586,0.0\n1.5175231527633284,0.22966493219687445
,1.0\n0.8958860863813638,-
1.1240959301759907,1.0\n2.0041894076557125,1.1916391957134091,1.0\n'}

```

```

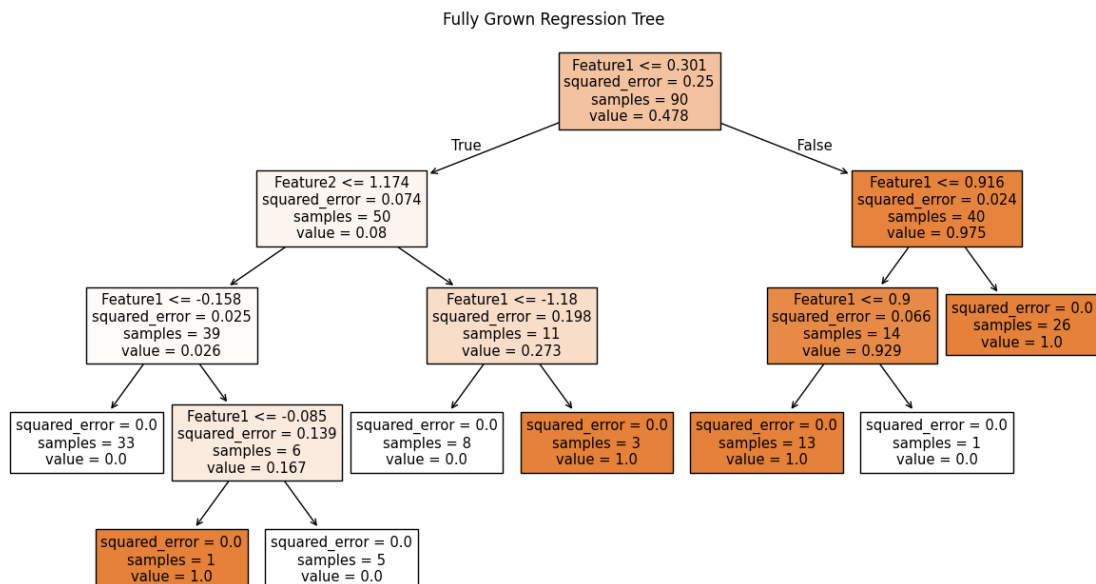
[ ]: import pandas as pd
from sklearn.tree import DecisionTreeRegressor, plot_tree
import matplotlib.pyplot as plt
data = pd.read_csv("A4_test_a4d66d34-6db0-44a4-a7c1-0aa26ab8d705.csv")

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

reg_tree = DecisionTreeRegressor(random_state=42)
reg_tree.fit(X, y)

plt.figure(figsize=(15, 8))
plot_tree(reg_tree, feature_names=X.columns, filled=True)
plt.title("Fully Grown Regression Tree")
plt.show()

```



Question 8

A.

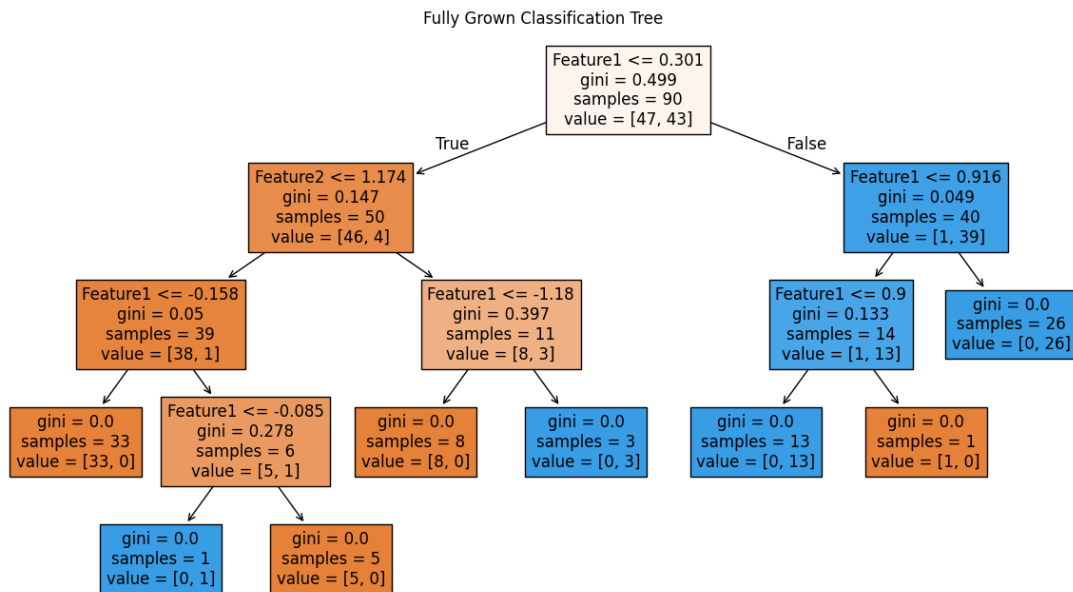
```
[ ]: from sklearn.tree import DecisionTreeClassifier

train = pd.read_csv("A4_test_a4d66d34-6db0-44a4-a7c1-0aa26ab8d705.csv")

X_train = train.iloc[:, :-1]
y_train = train.iloc[:, -1]

clf = DecisionTreeClassifier(criterion="gini", random_state=42)
clf.fit(X_train, y_train)

plt.figure(figsize=(15, 8))
plot_tree(clf, feature_names=X_train.columns, filled=True)
plt.title("Fully Grown Classification Tree")
plt.show()
```



B.

```
[ ]: import numpy as np
from sklearn.metrics import mean_squared_error

# Load test data
test = pd.read_csv("A4_train_74e601c5-f446-4f34-a53b-28ce05d1cec4 (1).csv")

X_test = test.iloc[:, :-1]
y_test = test.iloc[:, -1]
```

```

depths = range(1, 21)
sse_list = []

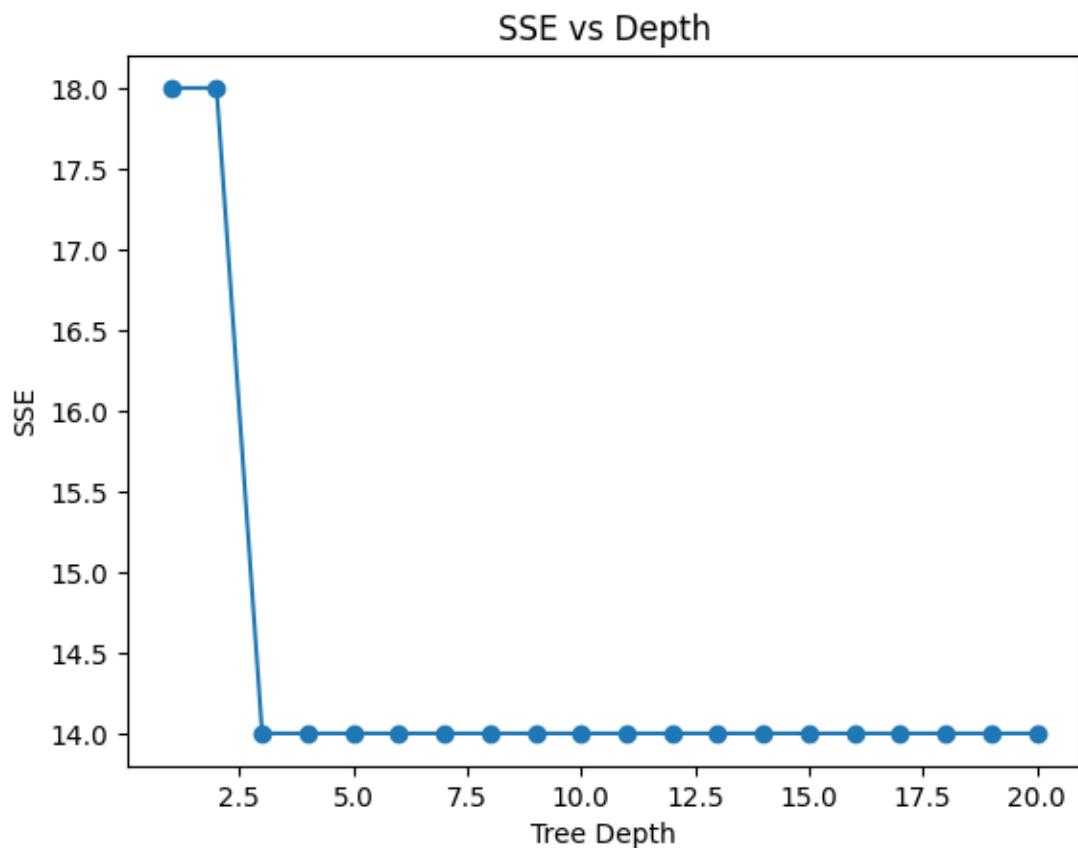
for d in depths:
    model = DecisionTreeClassifier(criterion="gini", max_depth=d,
    ↪random_state=42)
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    # SSE = sum of squared errors
    sse = np.sum((y_test - y_pred) ** 2)
    sse_list.append(sse)

plt.figure()
plt.plot(depths, sse_list, marker='o')
plt.xlabel("Tree Depth")
plt.ylabel("SSE")
plt.title("SSE vs Depth")
plt.show()

```



C.

```
[ ]: sse_diff = np.diff(sse_list)

    optimal_index = np.argmin(np.abs(sse_diff))

    optimal_depth = depths[optimal_index + 1]

    print("Optimal Depth:", optimal_depth)
```

Optimal Depth: 2

D.

```
[ ]: pruned_model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=optimal_depth,
    random_state=42
)

pruned_model.fit(X_train, y_train)

plt.figure(figsize=(15, 8))
plot_tree(pruned_model, feature_names=X_train.columns, filled=True)
plt.title(f"Pruned Tree (Depth = {optimal_depth})")
plt.show()
```

